

From Minimax to Blackwell Approachability

jungbinary.github.io

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Personal study notes compiled from various sources. They may contain errors. :)

Sion's minimax theorem

Definition 1 (Nash equilibrium in zero-sum game). Consider a two payer zero-sum game with action sets $\mathcal{A}$ and $\mathcal{B}$, and payoff function $f : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}$, where the learner minimizes f and nature maximizes f . A pair $(a^*, b^*) \in \mathcal{A} \times \mathcal{B}$ is a *Nash equilibrium* if

$$f(a^*, b) \leq f(a^*, b^*) \leq f(a, b^*), \quad \forall a \in \mathcal{A}, b \in \mathcal{B}.$$

Theorem 1 (Simplified form of Sion's minimax theorem). Let $\mathcal{A} \in \mathbb{R}^n$, $\mathcal{B} \in \mathbb{R}^m$ be compact convex sets. Let $f(a, b)$ be continuous, convex in a for a fixed b , and concave in b for a fixed a . Then,

$$\exists v \quad \text{s.t.} \quad \min_{a \in \mathcal{A}} \max_{b \in \mathcal{B}} f(a, b) = \max_{b \in \mathcal{B}} \min_{a \in \mathcal{A}} f(a, b) = v.$$

Proof. Weak duality is immediate. Thus, it is enough to show that

$$\min_{a \in \mathcal{A}} \max_{b \in \mathcal{B}} f(a, b) \leq \max_{b \in \mathcal{B}} \min_{a \in \mathcal{A}} f(a, b).$$

Suppose that a player chooses a_t according to a no-regret algorithm, while b is always chosen as $\arg \max_{b \in \mathcal{B}} f(a_t, b)$. Let the average strategies be

$$\bar{a} = \frac{1}{T} \sum_{t=1}^T a_t, \quad \bar{b} = \frac{1}{T} \sum_{t=1}^T b_t.$$

Now we bound the value of the min-max problem as

$$\begin{aligned} \min_{a \in \mathcal{A}} \max_{b \in \mathcal{B}} f(a, b) &\leq \max_{b \in \mathcal{B}} f(\bar{a}, b) \\ &\leq \frac{1}{T} \max_{b \in \mathcal{B}} \sum_{t=1}^T f(a_t, b) \quad (\because \text{Convexity}) \\ &\leq \frac{1}{T} \sum_{t=1}^T f(a_t, b_t) = \frac{1}{T} \left(\sum_{t=1}^T f(a_t, b_t) - \min_{a \in \mathcal{A}} \sum_{t=1}^T f(a, b_t) \right) + \frac{1}{T} \min_{a \in \mathcal{A}} \sum_{t=1}^T f(a, b_t) \\ &\leq \frac{\mathcal{R}_T}{T} + \frac{1}{T} \min_{a \in \mathcal{A}} \sum_{t=1}^T f(a, b_t) \\ &\leq \frac{\mathcal{R}_T}{T} + \min_{a \in \mathcal{A}} f(a, \bar{b}) \quad (\because \text{Concavity}) \\ &\leq \frac{\mathcal{R}_T}{T} + \max_{b \in \mathcal{B}} \min_{a \in \mathcal{A}} f(a, b) \end{aligned}$$

Take $T \rightarrow \infty$ on both side yields the desired result. $\square$

Blackwell approachability

Definition 2 (Blackwell approachability). Let $d(u, \mathcal{S}) = \inf_{z \in \mathcal{S}} \|u - z\|_2$ be the Euclidean distance between a point $u \in \mathbb{R}^d$ and $\mathcal{S}$. The target set $\mathcal{S}$ is approachable if there exists a strategy $\mathcal{A}$ of the learner such that for all sequences of nature's actions $\{b_t\}_{t \geq 1}$ with $b_t \in \mathcal{B}$ for all $t \geq 1$, using $\mathcal{A}$ to select the sequence of actions $\{a_t\}_{t \geq 1}$ with $a_t \in \mathcal{A}$ for all $t \geq 1$, ensures that

$$d\left(\frac{1}{T} \sum_{t=1}^T f(a_t, b_t), \mathcal{S}\right) \rightarrow 0 \quad \text{as } T \rightarrow \infty.$$

Theorem 2 (Blackwell's theorem). Let $\mathcal{S}$ be a closed convex set of $\mathbb{R}^d$ with $\|z\| \leq R$ for all $z \in \mathcal{S}$. Let $f : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}^d$ be a biaffine vector payoff over compact convex action sets with $\|f(a, b)\| \leq R$ for all $a \in \mathcal{A}, b \in \mathcal{B}$. If for all $b \in \mathcal{B}$, there exists $a \in \mathcal{A}$ such that $f(a, b) \in \mathcal{S}$, then $\mathcal{S}$ is approachable. Moreover, there exists a strategy such that

$$d\left(\frac{1}{T} \sum_{t=1}^T f(a_t, b_t), \mathcal{S}\right) \leq \frac{2R}{\sqrt{T}}.$$

Proof. For any fixed $w \neq 0$, by the Cauchy–Schwarz inequality,

$$|\langle w, z \rangle| \leq \|w\| \|z\| \leq \|w\| R, \quad \forall z \in \mathcal{S}.$$

Thus, the set $\{\langle w, z \rangle : z \in \mathcal{S}\}$ is bounded above, so $c := \sup_{z \in \mathcal{S}} \langle w, z \rangle$ is finite. By the definition of the supremum, $\langle w, z \rangle \leq c$ for all $z \in \mathcal{S}$. Therefore, we have a half space containing $\mathcal{S}$:

$$H := \{x \in \mathbb{R}^d : \langle w, x \rangle \leq c\}.$$

By assumption, for all $b \in \mathcal{B}$, there exists $a \in \mathcal{A}$ such that $f(a, b) \in H$. That is, for all $b \in \mathcal{B}$, there exists $a \in \mathcal{A}$ such that $\langle w, f(a, b) \rangle \leq c$, or

$$\max_{b \in \mathcal{B}} \min_{a \in \mathcal{A}} \langle w, f(a, b) \rangle \leq c.$$

Since f is biaffine, by Sion's theorem (what we've showed above),

$$\min_{a \in \mathcal{A}} \max_{b \in \mathcal{B}} \langle w, f(a, b) \rangle \leq c.$$

Since the above argument holds for any half-space H containing $\mathcal{S}$, we choose H_t so that the next payoff $f(a_t, b_t)$ pulls the running average $\bar{f}_t$ toward $\mathcal{S}$.

Let $\pi_t := \Pi_{\mathcal{S}}(\bar{f}_{t-1})$ be the Euclidean projection of $\bar{f}_{t-1}$ onto $\mathcal{S}$. We then consider the hyperplane passing through π_t with normal vector $\bar{f}_{t-1} - \pi_t$:

$$W := \left\{x \in \mathbb{R}^d : \langle x - \pi_t, \bar{f}_{t-1} - \pi_t \rangle = 0\right\}.$$

The corresponding half-space containing $\mathcal{S}$ is

$$H := \left\{x \in \mathbb{R}^d : \langle x - \pi_t, \bar{f}_{t-1} - \pi_t \rangle \leq 0\right\}.$$

This choice separates the current average $\bar{f}_{t-1}$ from $\mathcal{S}$ and ensures that the next payoff is chosen on the side of the hyperplane containing $\mathcal{S}$. And let a_H^* be an action such that $f(a, b) \in H$ for all $b \in \mathcal{B}$. Now,

by the definition of π_{t+1} , we have:

$$\begin{aligned}
d(\bar{f}_t, \mathcal{S})^2 &= \|\bar{f}_t - \pi_{t+1}\|^2 \\
&\leq \|\bar{f}_t - \pi_t\|^2 \\
&= \left\| \left(\frac{t-1}{t} \bar{f}_{t-1} + \frac{1}{t} f_t \right) - \pi_t \right\|^2 \\
&= \left\| \left(\frac{t-1}{t} (\bar{f}_{t-1} - \pi_t) + \frac{t-1}{t} \pi_t + \frac{1}{t} f_t \right) - \pi_t \right\|^2 \\
&= \left\| \frac{t-1}{t} (\bar{f}_{t-1} - \pi_t) + \frac{1}{t} (f_t - \pi_t) \right\|^2 \\
&= \left(\frac{t-1}{t} \right) d(\bar{f}_{t-1}, \mathcal{S})^2 + \frac{\|f_t - \pi_t\|^2}{t^2} + 2 \frac{t-1}{t^2} \langle f_t - \pi_t, \bar{f}_{t-1} - \pi_t \rangle.
\end{aligned}$$

Since $f_t \in H$, the last term is negative. Since f_t and π_t are both bounded by R , The middle term is bounded by $4R^2/t^2$. ($\because \|f_t - \pi_t\| \leq \|f_t\| + \|\pi_t\|$, thus $\|f_t - \pi_t\|^2 \leq (\|f_t\| + \|\pi_t\|)^2 = \|f_t\|^2 + \|\pi_t\|^2 + 2\|f_t\|\|\pi_t\| \leq 4R^2$). Letting $\zeta_t^2 = t^2 d(\bar{f}_t, \mathcal{S})^2$, we can obtain a recurrence relation

$$\zeta_t^2 \leq \zeta_{t-1}^2 + 4R^2.$$

Rearranging this terms gives the desired bound. □

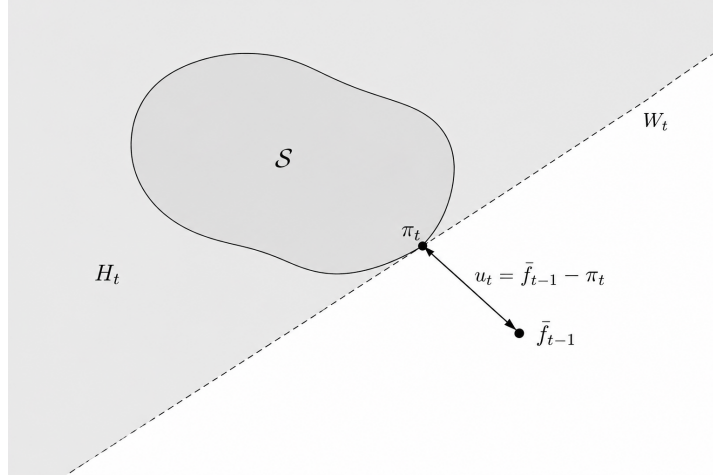


Figure 1: Visualization of Blackwell's strategy

Reading list

- Shie Mannor and Nahum Shimkin, *A Geometric Approach to Multi-Criterion Reinforcement Learning*, Journal of Machine Learning Research (JMLR), 2004.
- Brian W. Lee, Nika Haghtalab, Michael I. Jordan, and Ryan J. Tibshirani, *Blackwell Approachability and Gradient Equilibrium are Equivalent*, Conference on Learning Theory (COLT), 2026.
- Christian Kroer, *Games, Markets, and Online Learning*, Cambridge University Press, 2026.